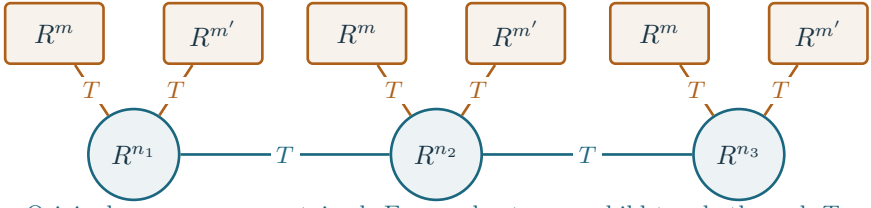
Adding long leaf pairs to a knot core

A tree of ordered tangle operations; one added pair is shown at each vertex.



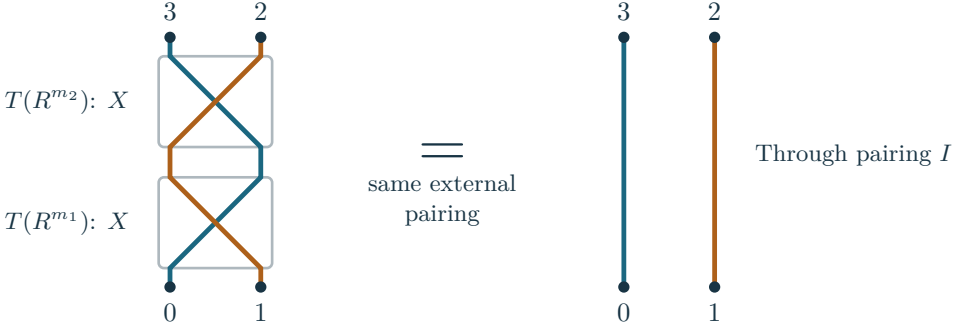
Original powers n_v are retained. Every edge turns a child tangle through T.

At each original vertex add $B \geq 2^{28}$ leaves, with B even.

Each added power is odd and has $|m| \geq 2^{27} + 1$.

What one added pair does to the endpoints

Only strand connections are drawn here; over and under crossings are forgotten.



$X = \{02, 13\}, \qquad I = \{03, 12\}, \qquad X\,X = I.$

Insert each even stack in one specified product slot, preserving the cyclic order.

This preserves the knot pairing; it does not remove the added braids by an isotopy.